

#	Project Name	Team members	Project Description	DEMO	Due Date
				5/13/2022	5/17/2022
1	Smart Robot Car (Arduino)	Minh Nhat Duong Nguyen Do Anh Le	This project involves creating a smart cart that can move forward, backward, and turn left or right, as well as stop. We will include a front sensor that detects obstacles. If an obstacle is too close to the cart, it will automatically turn to the side or make a U-turn to avoid it. Additionally, we...have black tape on the ground to guide the way the cart should go. Coding is required for this project written in Arduino compiler to link to the project. Otherwise, we can practice building the cart by connecting separate parts from the buying item.		
2	Smoke Detector (Arduino)	Jassan Dhami Nhan Nguyen	This program detects smoke in the area and displays a specific color combination based on the amount of particulate matter detected by the sensor. Additionally, a buzzer sounds to alert the user of nearby smoke. We will construct the smoke detector using a breadboard, wires, the sensor, the Arduino, and a speaker. We chose this project because we find it intriguing to have a device that can detect particulate matter and output a corresponding voltage. It will also provide us with valuable experience in working with Arduino and breadboards.		
3	The fan regulates the temperature (Arduino)	Abdulrahman Abuammash Pramodya Rajapakse	The fan speed can be controlled according to temperature and humidity using an electronic circuit using the Arduino board. The Arduino board is currently very progressive among all electronic device circuits, so we used an Arduino board for the fan speed. The proposed system is designed to detect room temperature and send this information to the Arduino board. Then the Arduino board implements the contrast from the current temperature and adjusts the temperature based on the built-in Arduino software. The result obtained from the operation is transmitted by the o/p port of the Arduino board to the relevant LCD data. Board-generated pulses that feed the driver circuit for fan-favorite output.		
4	The Parking Counter (Arduino)	Hamza Ahmed Anthony La	The Arduino kit is designed to create a parking lot that uses props, such as boxes, to represent simple or toy cars. The parking lot will feature an entrance with a gate that raises or lowers based on the available capacity inside. The system operates with a pre-programmed maximum capacity that corresponds to the number of cars the parking lot can accommodate. When a car enters, the counter decreases by one; when a car leaves, the counter increases by one. By default, the gate remains down and lifts only if the counter is greater than zero, indicating that there is an available parking spot. Additionally, the parking lot will use sensors to track the number of cars in each section (currently planned for two or three sections). A small display will show which section has the most available spots. The system will comprise two classes or structures: ParkingGate, which controls the gate, and ParkingLot, which encompasses all the functions and variables needed to manage capacity effectively.		
5	Smart car with obstacle avoidance, light, line and color tracing (Raspberry Pi)	David Yu HongYe Zheng Su Aye	Smart car with obstacle avoidance, light tracing, line tracing, colorful light, ultrasonic camera. The car runs on raspberry and has 4 functions. The obstacle avoidance function of the car mainly uses HC-SR04 ultrasonic module, which is installed with the servo. The ultrasonic module measures the obstacle distance in the left, middle and right directions. The car performs corresponding actions according to different distances. The line tracking function of the car mainly uses infrared sensors. When the sensors detect a black line, the corresponding led will light up. The car performs corresponding actions according to the values of three sensors.		

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Security Camera (Raspberry Pi)	Ryan Ross Thomas Sweat	We are planning to create a security camera system using a Raspberry Pi and a camera module to monitor for any animals that may harm our trees. The camera will be positioned near or even within the trees, allowing it to survey the surrounding environment for any potential wildlife threats. If the camera detects sufficient movement, it will trigger an alarm to scare off any wild animals or at least alert us so we can address the situation manually. Additionally, we will have a monitor displaying the live feed from the camera.		
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Robot Vacuum (Raspberry Pi)	Ying Zhou	For the programming project, I plan to design a robot vacuum with the following support features: 1. Manual mode. Owner uses a remote controller to move the robot. 2. Auto mode. Robot vacuum will clean all empty spaces and avoid any obstacles. 3. Read map for a csv file. Map will be defined as a two dimensional matrix. Value of each cell will be 0:Empty, 1:Obstacle 4. Auto generate map given #obstacles. 5. Print the cleaning path Robot can 1. Move forward. Move 1 step forward 2. Turn left. Counter clockwise 90 degrees. 3. Turn right. Clockwise 90 degrees. 4. Detect obstacles. The front sensor will detect if there is any obstacle in the front direction. Remote controller can		
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Detect Motion (Raspberry Pi)	Christopher Wan and Evan Brannon-Wu	Our motion sensor detects movement and can activate a light or alarm. It tracks two states: motion (1) and no motion (0). When there's no motion (0), the system remains inactive. If motion is detected (1), it will turn on the light or sound the alarm, depending on its placement near the room's entrance. The device also monitors temperature and adjusts the number of lights through smart plug integration based on how many people are present.		
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LED circuit to make the LEDs blink (Raspberry Pi)	Abdarrahman Ayyaz Jotsaroop Singh	Our project involves an LED board where we alternate between different states to light up the LEDs. This helps us understand how to control specific areas of our circuit and how to organize the various components in both the code and the circuit at the same time. The specific details will depend on the kits we receive and the number of LEDs available, although we already have an idea of which states to include.		
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Motion Sensor (Arduino)	Brandon Villiados and Niko Mcniel	Our group has decided to work with a motion sensor. Our original thought was to use a motion sensor with an Arduino and program it to set off an alarm whenever motion was sensed within a certain distance, however, we decided to work with light instead because we think that we can customize lights to do a lot more things than just an alarm. We will use the Arduino PIR motion sensor to detect infrared signals that can be detected from human movements. We haven't decided what kind of LEDs we would use to indicate that the motion sensor detected movement but what we aim to do is light a different LED depending on the distance or movement detected by the sensor.		
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Smart water hygiene (Arduino)	Arlyn Benoza - Kenneth Manuntag	This project is a hand wash timer designed to ensure that individuals wash their hands for an appropriate duration—not too short or too long. My partner and I are currently searching for materials in addition to the Arduino kit, which is expected to ship next week. If time allows, we may consider adding a music box or some type of audio alert to inform users when they should stop washing their hands.		
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T-cobblers - Guitar Hero (Raspberry Pi)	Kalp Patel Varun Narravula	We are developing a drumming game similar to Guitar Hero. Users will choose from three randomly generated patterns created from a truth table of their preferred size, with three rows indicating hits (1) and non-hits (rests) (0). Based on the input, a boolean expression will be generated. If the expression evaluates to 1, the pattern was played correctly; if it evaluates to 0, it was not.		
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RGB (Raspberry Pi)	Mahnoor Kafil Sadiyah Khan	Our project involves controlling a group of RGB LED strip lights. We will be able to control the lights' brightness, colors, and custom patterns. It will also be programmed to be turned off and on. We have not started compiling our code yet, but we plan on using an LED library and having a setup function to switch the LEDs on and off and a function to control the brightness which will form new colors. Arrays and variables will be used to store the pin numbers for the colors that the LEDs will flash and for loops to loop through the number of LEDs present in the strip and flash them with the commands. We plan on using binary numbers to control the brightness of the pixels by storing the numbers that correlate with the colors. The higher the binary number, the higher the brightness of the pixels. The brightness can be adjusted for each RGB color, to make new colors.		
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Smart Robot Car Kit V4 (Arduino)	Ngoc Huynh Bao Truong	For this project, we will need a few key components: the UNO R3 controller board, an LCD screen, and a 4x4 keypad. The keypad has eight pins—four for the rows and four for the columns. The program will determine which button is pressed by checking the row and column. It will then store the digit or operator linked to that button in a character array. When the “=” button is pressed, the program will perform the calculations and show the result on the LCD screen. A calculator is a key tool for students, and coding basic math functions is often one of the first programming tasks. We are building a simple calculator using Arduino that can add, subtract, multiply, and divide.		